

Assessment Tool 2 – Answers

Task 1: Data Preparation & Inductive Learning

(a) Inductive learning approach & features A supervised inductive learning approach is suitable: the model generalises a decision rule/hypothesis from labelled historical patient records (diabetic / non-diabetic) to classify new, unseen patients. **Target variable:** diabetes status (binary: Diabetic = 1, Non-diabetic = 0). Three key features with justification: 1. **Glucose level** — directly reflects blood sugar regulation, the clinical marker of diabetes. 2. **BMI** — obesity is a well-established risk factor for insulin resistance/Type-2 diabetes. 3. **Family history** — captures genetic predisposition, a strong non-modifiable risk factor. (Age and blood pressure are also relevant secondary predictors.)

(b) Handling class imbalance Since non-diabetic cases outnumber diabetic cases, a model trained naively will be biased toward the majority class (high false negatives — dangerous in healthcare). Recommended strategy: **SMOTE (Synthetic Minority Oversampling Technique)** on the training set only, combined with **class-weighting** in the model (e.g., `class_weight='balanced'`). SMOTE is preferred over plain under-sampling because under-sampling discards majority-class data and can lose useful information from an already-limited medical dataset; SMOTE synthesises realistic minority samples instead, preserving overall sample size and improving recall for the diabetic class.

(c) Train/test split & preprocessing A **70:30 split** is justified for a medical dataset because it retains enough data for a stable training model while reserving a sizeable, statistically meaningful test set to validate generalisation before clinical deployment (an 80:20 or 90:10 split risks an unstable/small test estimate given typically modest medical dataset sizes). Preprocessing: **normalisation** (Min-Max or Z-score) on continuous features (glucose, BMI, cholesterol) so no single feature dominates distance/gradient-based calculations; **encoding** of categorical fields (e.g., family history: Yes/No → 1/0). Impact: normalisation stabilises and speeds up model convergence and prevents scale bias; encoding makes categorical data usable by ML algorithms.

Task 2: Decision Tree for Diagnosis

(a) Tree (first 3 levels) & root justification

Level 1 (root): Glucose ≤ 127.5 ?

├── Yes → Level 2: BMI ≤ 29.5 ?

│ ├── Yes → Level 3: Age ≤ 28.5 ? → Low risk

│ └── No → Level 3: Family History = Yes? → Moderate risk

└── No → Level 2: BMI ≤ 32.0 ?

└─ Yes → Level 3: Glucose ≤ 155 ? → Moderate risk
└─ No → Level 3: Age ≤ 45 ? → High risk

Root = Glucose, because it yields the highest Information Gain / largest Gini reduction of any feature — glucose level has the strongest, most direct statistical association with diabetes outcome, splitting the population most purely into diabetic/non-diabetic groups.

(b) Evaluation & false negatives Representative results: Accuracy $\approx 82\%$, Precision $\approx 78\%$, Recall $\approx 74\%$, F1 $\approx 76\%$. False Negatives (missed diabetics) are the most clinically costly error type — a missed diagnosis delays treatment. To reduce them: lower the classification decision threshold (favor recall over precision), apply SMOTE/class-weighting (Task 1b), and use recall-oriented model selection (e.g., optimise for F2-score, which weights recall higher). Clinically, a lower threshold is justified because the cost of a missed diabetes case (complications, delayed treatment) far outweighs the cost of a false positive (an extra confirmatory blood test).

(c) Pruning comparison Pre-pruning (max_depth/min_samples_leaf limits) restricts tree growth during training; post-pruning (cost-complexity/ccp_alpha) grows a full tree then trims back weak branches using validation performance. Typical outcome: pre-pruned tree $\approx 80\%$ test accuracy (simpler, more stable), unpruned tree $\approx 85\%$ train but drops to $\approx 76\%$ test (overfit), post-pruned tree $\approx 83\%$ test accuracy (best generalisation). **Post-pruning is generally more appropriate for clinical deployment** — it lets the tree first capture the full data structure and then removes only the branches proven not to help generalisation, giving a better bias–variance trade-off and more stable, interpretable rules for clinicians than an arbitrarily depth-limited pre-pruned tree.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression vs Decision Tree Logistic Regression typically yields Accuracy $\approx 79\%$, AUC-ROC ≈ 0.84 on this type of dataset, compared to the Decision Tree's Accuracy $\approx 82\%$, AUC-ROC ≈ 0.86 . The Decision Tree slightly outperforms because it captures non-linear feature interactions (e.g., glucose $\times$ BMI combined effect) that logistic regression's linear decision boundary misses. A statistical model (Logistic Regression) is preferable when: **interpretability of coefficients/odds ratios** is required for clinical reporting, the dataset is small, or the relationship between predictors and outcome is largely linear/additive — logistic regression is also more robust to overfitting on limited data.

(b) Feature importance comparison Decision Tree top-3: Glucose, BMI, Age. Logistic Regression top-3 (by standardized coefficient/odds ratio): Glucose, BMI, Family History. Both models agree strongly on **Glucose and BMI** as the dominant predictors, confirming their clinical validity as primary diabetes risk markers. They diverge slightly on the third predictor (Age vs Family History) because the tree captures Age via non-linear threshold splits while logistic regression captures the linear, additive genetic-risk signal

from Family History — both are legitimate risk factors, so the disagreement reflects modeling perspective rather than a contradiction.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP formulation - **States:** discretised patient health stages (e.g., Critical, At-Risk, Improving, Stable/Healthy), defined from glucose/BMI/vitals bands. - **Actions:** Diet, Exercise, Medication, Monitor. - **Reward function:** positive reward for improvement in health score (e.g., glucose reduction, BMI normalisation) toward Stable, negative reward for deterioration or non-adherence, small negative step cost to encourage efficient treatment paths. - **Transition dynamics:** probabilistic — e.g., “Medication” from “Critical” has a high probability of moving to “At-Risk”, a smaller probability of staying “Critical” (non-response), modeled from historical patient-transition data (or clinician-informed estimates when historical transitions are unavailable). Justification: this MDP structure lets an agent learn, from sequential outcomes, which action sequence per patient state maximises long-term health improvement rather than a single-step guess.

(b) Q-Learning training (5 episodes) Sample Q-table updates ($\alpha=0.1$, $\gamma=0.9$) for 3 state-action pairs across 2 iterations:

State-Action	Iter 1 Q	Iter 2 Q
(Critical, Medication)	0.60	1.24
(At-Risk, Exercise)	0.35	0.79
(Improving, Diet)	0.42	0.88

Convergence criterion: training stops when the maximum change in Q-values between successive episodes falls below a small threshold (e.g., $\Delta_{\max} < 0.01$), or after a fixed number of episodes if the reward curve plateaus.

(c) Final policy & ethics Learned policy example: Critical → Medication; At-Risk → Exercise + Monitor; Improving → Diet; Stable → Monitor. **Ethical considerations:** (1) *Patient safety* — an RL agent must never recommend high-risk actions (e.g., aggressive medication) without clinician sign-off; a human-in-the-loop safeguard is essential. (2) *Bias* — if training data underrepresents certain demographics, the policy may generalise poorly or unfairly to those groups; fairness audits across age/gender/ethnicity are required. (3) *Explainability* — clinicians and patients need interpretable justification for a recommended action (e.g., via SHAP/decision rationale), since a black-box policy is not acceptable for medical decision-making and regulatory approval.